INTERNSHIP REPORT

Submitted In Partial Fulfilment of Requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

Submitted to

DEENBANDHU CHOOTU RAM UNIVERSITY OF SCIENCE AND TECHNOLOGY, MURTHAL, SONEPAT

by

AMIT YADAV (22001001013)

Under the Supervision of

Dr. Neetu Verma

(Professor, CSE Department)

Faculty of Engineering and Technology

DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY

DECLARATION

I hereby declare that the project report entitled **"SUREFY.AI"** submitted to **Deen Bandhu Chhotu Ram University of Science and Technology, Murthal** is an authentic record of my own work carried out during the academic year 2025–2026 under the guidance of my project mentor. This work has not been submitted previously, either in part or full, to this or any other university or institution for the award of any degree or diploma.

I take full responsibility for the originality of this project and confirm that all sources used have been duly acknowledged.

Signature of students:

AMITYADAV

(22001001013)

Place: DCRUST, Murthal

Date:

DEENBANDHU CHHOTU RAM UNIVERSITY OF SCIENCE AND TECHNOLOGY, MURTHAL, SONEPAT, HARYANA-131039

Department of Computer Science and Engineering

This is to certify that the Internship Report submitted by **AMIT YADAV (Enrollment No. 22001001013)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of **Deenbandhu Chhotu Ram University of Science and Technology, Murthal**, for the internship carried out **at SUREFY.AI(Shankar groups)** for a duration of **6 months (from 18th Nov 2025 to 28th May 20256)** is the original work carried out by him under my supervision and guidance.

Prof (Dr.) NEETU VERM

(Department Supervisor)

This is to certified that the B.Tech External Internship viva-voice Examination of AMIT YADAV(22001001013) has been held on dated …./…../20…. and accepted the internship for the award of the degree.

Dr. Ajmer Singh

Internship coordinator

External Examiner **Chairperson (CSED)**

CERTIFICATION

SANKAR GROUP

INTERNSHIP CERTIFICATE

This is to certify that

Amit Yadav

BAWAL ROAD , REWARI , HARYANA 123401

has successfully completed his internship as "Frontend Developer Intern" with **Sankar Group** from **18 Nov 2024** to **18 May 2025**.

During the internship period, Amit was involved in various tasks and projects related to frontend development. He demonstrated a good understanding of concepts, great enthusiasm, and a strong willingness to learn.

We were satisfied with his performance and found him to be sincere, hardworking, and a valuable addition to the team.

We wish Amit all the best for his future endeavors.

With regards,

Sankar Rao Vysyaraju
CEO & Founder

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ABSTRACT

Surefy.ai ReachOut is a web-based lead management application developed to help businesses efficiently capture, organize, and track customer leads while automating communication and follow-up processes. In many organizations, lead handling is still performed manually using spreadsheets, calls, or scattered tools, which often results in missed opportunities, delayed responses, and poor customer engagement. Surefy ReachOut addresses these challenges by providing a centralized, secure, and easy-to-use platform accessible through any web browser.

The system is built using modern web technologies, including the MERN stack—MongoDB for database management, Express.js and Node.js for backend services and API handling, and React.js for creating a dynamic and responsive user interface. The platform supports role-based access, allowing administrators to manage leads, campaigns, reports, and team activities, while users can handle contacts, communicate via integrated WhatsApp chat, and track lead progress efficiently.

Throughout the development of the project, emphasis was placed on creating a streamlined workflow—from lead capture through forms to conversion tracking and reporting. The system includes features such as real-time lead status updates, campaign management, contact organization, and performance analytics. These features improve response time, enhance productivity, and ensure better data organization compared to traditional methods.

Developed as a practical solution for modern businesses, Surefy ReachOut not only improves lead management efficiency but also demonstrates the application of full-stack development concepts, including API integration, real-time data handling, and scalable system design.

ACKNOWLEDGMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the development of this project.

First, I thank my project guide, **Mrs. Neetu verma**, for his continuous guidance, valuable suggestions, and encouragement during the entire project. His expert advice helped me overcome many challenges and improve the quality of my work.

I extend my thanks to the Head of the Department, **Mr. Sukhdeep Singh**, for providing me the opportunity and necessary resources to carry out this project.

Finally, I thank all my friends and family for their motivation and support throughout this journey.

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INTRODUCTION OF COMPANY

Surefy.ai is a fast-growing technology platform focused on providing intelligent solutions for lead management, customer communication, and business automation. The company specializes in developing modern web-based applications that help organizations streamline their sales processes, manage customer interactions, and improve overall operational efficiency.

Founded with the vision of simplifying business workflows, Surefy.ai aims to bridge the gap between traditional lead handling methods and advanced digital systems. Its mission is to empower businesses with smart tools that enhance productivity, improve response time, and increase conversion rates through automation and real-time data management.

Surefy.ai promotes a culture of innovation, continuous improvement, and user-centric design. The platform integrates modern technologies and scalable architectures to deliver reliable and efficient solutions. It focuses on building systems that are easy to use, highly responsive, and adaptable to the evolving needs of businesses.

The company serves a wide range of clients, including startups, small and medium enterprises (SMEs), and growing organizations that require efficient lead handling and communication systems. Its solutions cover the complete workflow—from lead capture and tracking to communication, reporting, and performance analysis—ensuring a seamless business experience.

In addition to its core services, Surefy.ai also supports learning and development by providing opportunities for students and interns to gain practical experience in real-world projects. Through hands-on training and exposure to modern technologies, the platform contributes to building skilled professionals in the field of web development and business automation.

INTRODUCTION OF SUREFY .AI

1.1 Project Overview

Surefy.ai ReachOut is a web-based lead management and communication platform developed using modern technologies such as the MERN stack — MongoDB, Express.js, React.js, and Node.js. The main purpose of the system is to help businesses efficiently capture, manage, and track customer leads while automating communication and follow-up processes.

The platform provides a centralized dashboard where users can monitor lead status, manage contacts, run campaigns, and communicate with customers through integrated tools like WhatsApp chat. Admin users can manage leads, campaigns, reports, and team activities, while other users can handle daily lead interactions and updates. The system is designed with role-based access control, ensuring that sensitive data is accessed only by authorized users.

This project focuses on creating a user-friendly, scalable, and secure solution that replaces traditional manual lead handling methods such as spreadsheets and disconnected tools, improving efficiency and business productivity.

1.2 Problem Statement

In many businesses, lead management is still handled manually through spreadsheets, phone calls, or multiple unconnected platforms. This approach often leads to data duplication, missed follow-ups, delayed responses, and poor tracking of customer interactions.

Without a centralized system, it becomes difficult to manage lead status, monitor team performance, and maintain proper communication with potential customers. Lack of automation also results in inefficiency and reduced conversion rates.

The goal of this project is to develop a secure and easy-to-use web application that centralizes lead management, enables real-time tracking, improves communication, and ensures role-based access control for better data security and workflow management.

1.3 Feasibility Study

Technical Feasibility:

The system is built using the MERN stack, which is widely used, scalable, and supported by a large developer community. MongoDB provides flexible data storage for leads and contacts, while React.js ensures a fast and responsive user interface. Node.js and Express.js enable efficient backend processing and API development. These technologies allow smooth integration of features like real-time updates, dashboards, and communication tools.

Operational Feasibility:

The platform is designed with a simple and intuitive interface, making it easy for users to manage leads without extensive training. Admins can control and monitor system activities through dashboards, while users can easily handle leads, contacts, and communication tasks. Being web-based, the system is accessible from any device with an internet connection.

Economic Feasibility:

The project uses open-source technologies, which reduces development and licensing costs. It can be deployed on affordable cloud platforms with minimal hardware requirements. This makes the system cost-effective and suitable for businesses of all sizes.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

- Personal computer or laptop with at least 4 GB RAM.
- Minimum 50 GB hard disk space for software installation and data storage.
- Reliable internet connection for development and deployment.

Software Requirements

- **Operating System:** Windows, Linux, or macOS.
- **Node.js and npm:** Runtime environment and package manager for backend development.
- **Express.js:** Web application framework to create APIs.
- **React.js/ Next.js:** Frontend library for building user interfaces.
- **Visual Studio Code:** Code editor used for development.
- **Postman:** Tool to test backend API endpoints.
- **Git & GitHub:** Version control and repository hosting for team collaboration.

TOOLS AND TECHNOLOGY USED

- **MongoDB**
  - Why chosen: Flexible NoSQL database that handles complex and varying data easily. ○ Features: Document-based storage, scalability, high performance. ○ Uses in Project: Stores employee details, leave records, and user roles securely.
- **Next.js**
  - Why chosen**:** Simplifies full-stack development by combining frontend and backend capabilities in a single framework with built-in routing and performance optimization.

    • **Features:** Server-side rendering (SSR), static site generation (SSG), API routes, fast performance, and SEO-friendly architecture.

    • **Uses in Project:** Builds dynamic user interfaces, handles routing, manages server-side rendering, and supports API endpoints for seamless data handling.
- **React.js**
  - Why chosen: Creates dynamic, fast, and reusable user interface components. ○ Features: Component-based architecture, virtual DOM for fast rendering, state management. ○ Uses in Project: Builds admin and employee dashboards, login pages, and leave application forms.
- **Node.js**
  - Why chosen: Allows running JavaScript on the server for full-stack development. ○ Features: Event-driven, non-blocking I/O, scalable backend environment. ○ Uses in Project: Runs backend server, processes API calls, connects frontend with database.

- **Git and GitHub**
  - Why chosen: Version control system for efficient code management and collaboration. ○ Features: Tracks changes, branches, merges, remote repository hosting. ○ Uses in Project: Maintains project code versions and backups throughout development.
- **Postman**
  - Why chosen: Easy and effective tool for API testing and debugging. ○ Features: Send various HTTP requests, inspect responses, automate tests. ○ Uses in Project: Tests backend APIs to ensure correct functionality before frontend integration.
- **Visual Studio Code**
  - Why chosen: Powerful and lightweight code editor with rich extensions. ○ Features: Syntax highlighting, debugging, version control integration. ○ Uses in Project: Write, edit, and debug frontend and backend code efficiently.

IMPLEMENTATION OF SUREFY.AI

1. Frontend (Next.js)

• **Created using:** Next.js (React-based framework)
• **Folder Structure:**
o **components/** – Reusable UI components (Navbar, Sidebar, Cards, Modals)
o **app/** or **pages/** – Different views (Dashboard, Leads, Campaign, Reports, etc.)
o **store/** / **redux/** – State management (user data, lead data, auth state)
o **services/** – API integration using Axios or Fetch
o **utils/** – Helper functions and configurations
o **layout/** – Common layout (Sidebar + Header structure)
o **middleware/** – Route protection and authentication handling

2. Backend (Node.js + Express.js)

• **Initialized using:** npm init
• **Folder Structure:**
o **routes/** – API route handlers (auth, leads, campaign, contacts, reports)
o **controllers/** – Business logic for each module
o **models/** – Mongoose schemas (User, Lead, Contact, Campaign, Ticket)
o **middlewares/** – Authentication, authorization, and validation
o **config/** – Database and environment configurations
o **server.js / index.js** – Entry point of backend server

3. Database (MongoDB)

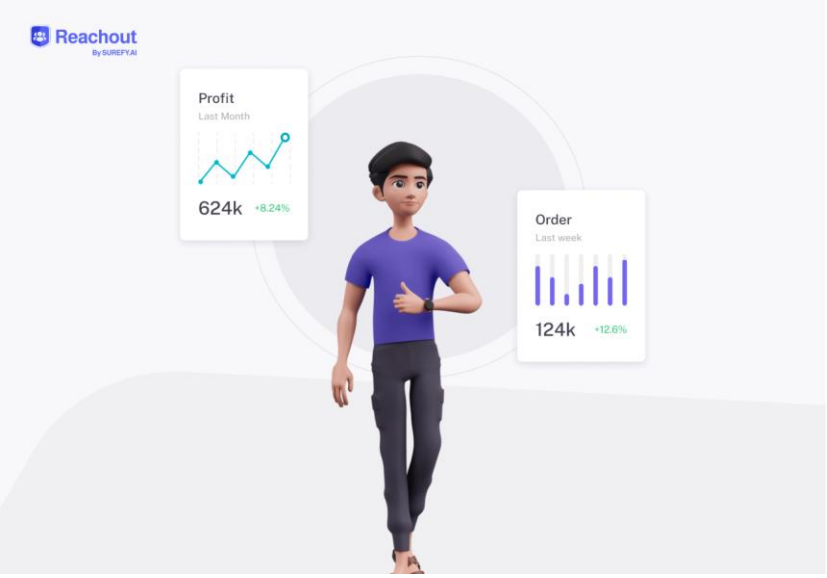
• **Hosted on:** MongoDB Atlas (cloud-based database)
• **Data Handling:**
o Stores data in flexible JSON document format

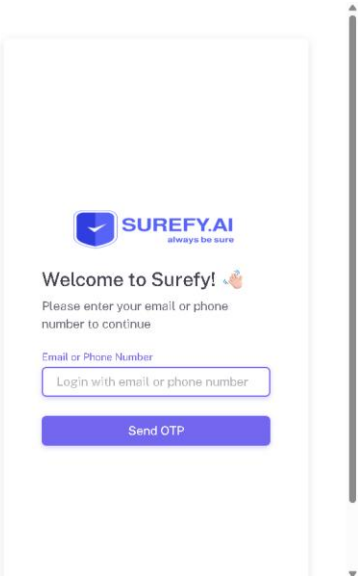
o Collections include Users, Leads, Contacts, Campaigns, Tickets, Reports
o Provides fast querying and scalability for real-time lead tracking

Core Functionalities -

1. Authentication & Authorization

• **Used:** JWT (JSON Web Token) for session management, bcrypt for secure password/OTP handling
•**KeyFeatures:**
o Login using Email or Phone Number with OTP verification
o Tokens stored securely (localStorage/cookies) and verified on each request
o Role-based access control (Admin, Team Member, etc.)
o Protected routes using middleware to restrict unauthorized access





2. Lead Management

- **Admin/User can:**

o Add new leads manually or through integrated forms
o Update lead details (name, contact, status, notes)
o Delete or archive leads
o Track lead stages (New, Contacted, Warm, Hot, Not Interested, Closed)
o Assign leads to team members

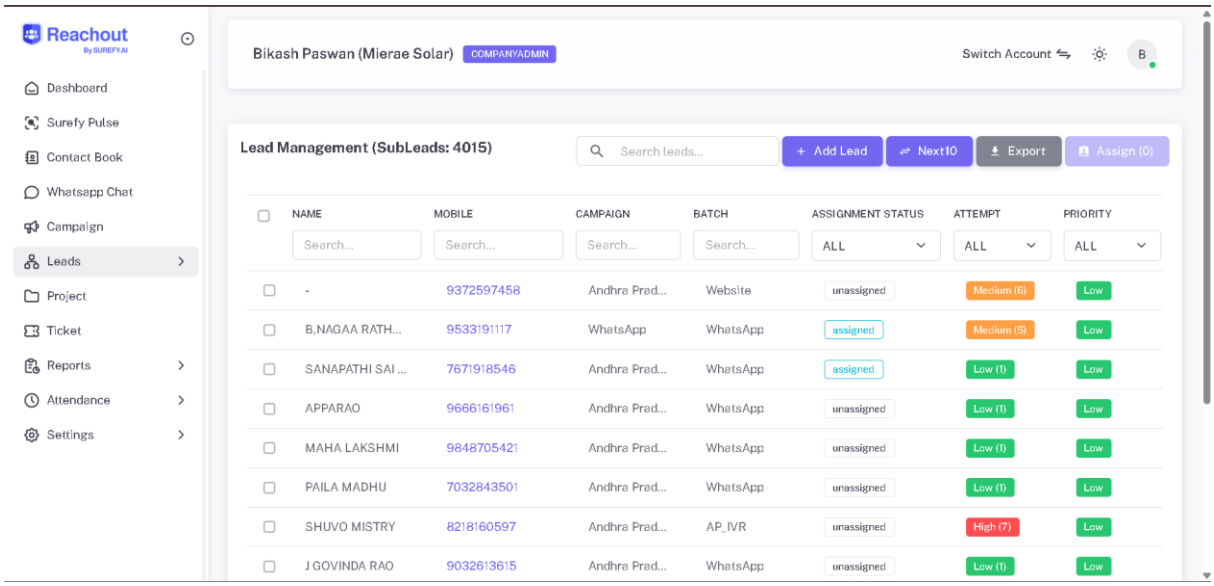


Fig . 2.1

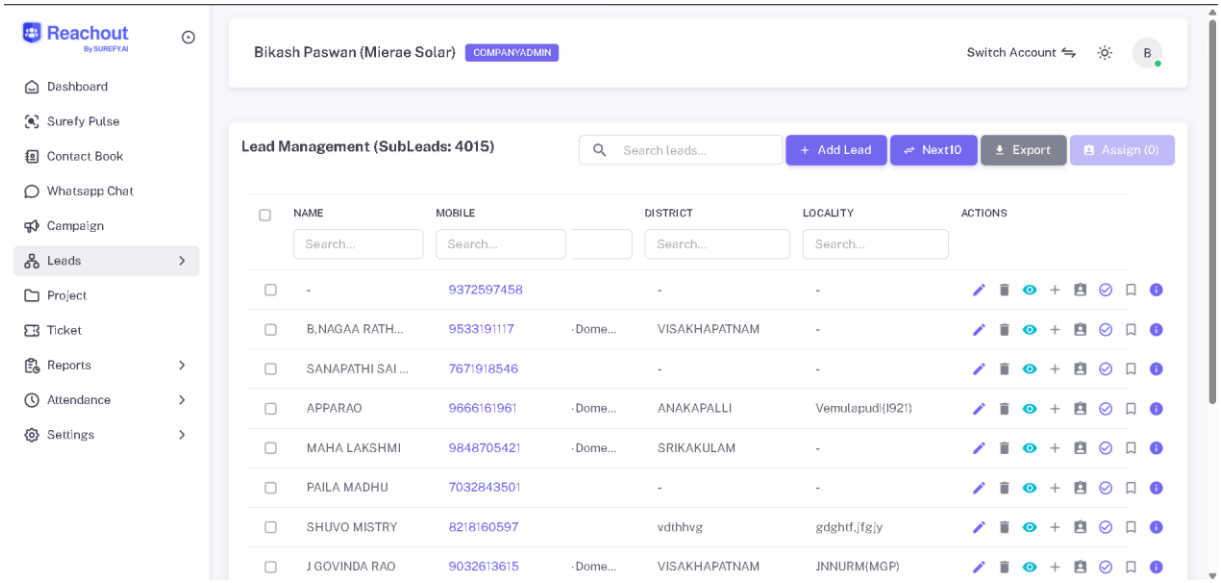


Fig . 2.2

3. Dashboard & Analytics

- Features:

o Displays real-time statistics (Call Status, Lead Status)
o Lead summary based on categories (Hot, Warm, Cold, etc.)
o Performance tracking of team members
o Filters by date (Today, All, etc.)

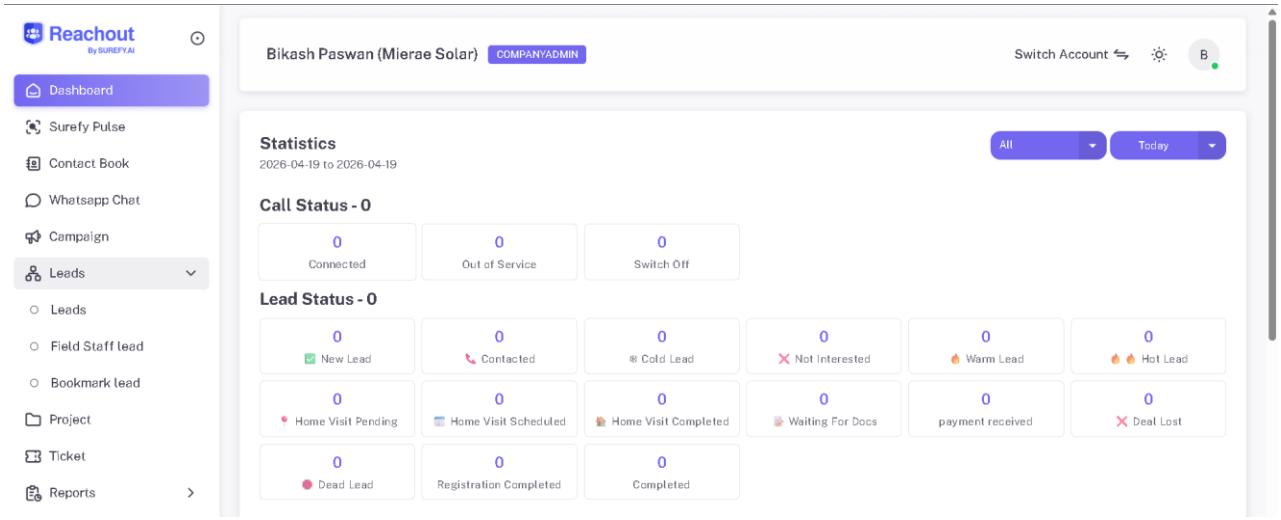


Fig . 3.1

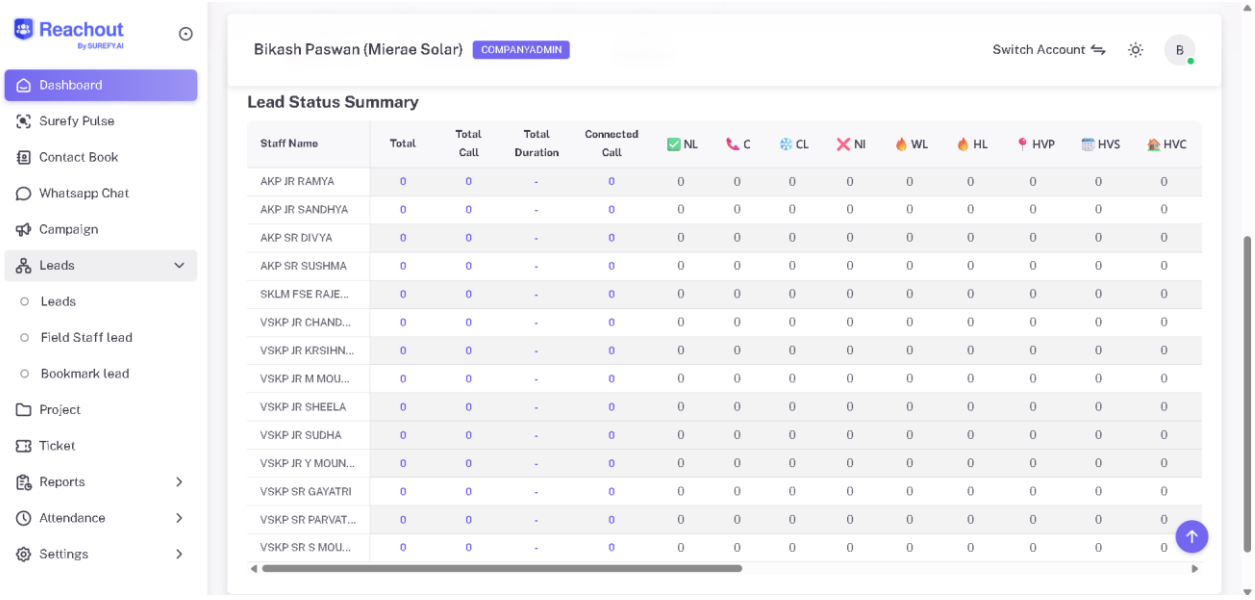


Fig . 3.2

4. Contact Book Management

- User can:

o Store and manage all customer contacts
o Edit and update contact details
o Use contacts for communication and campaigns

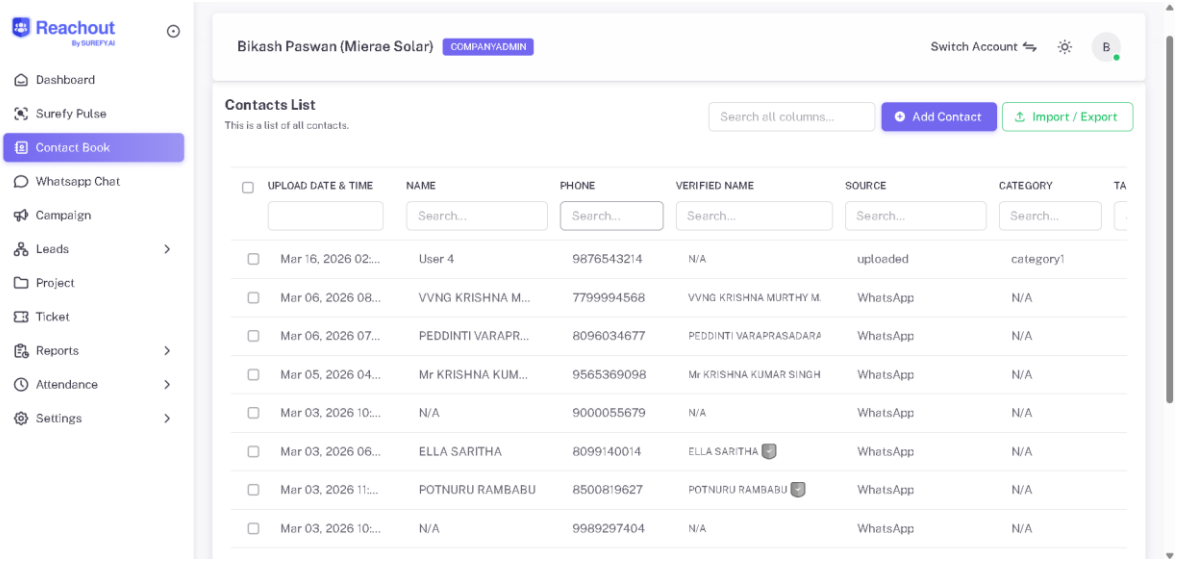


Fig . 4.1

5. Communication (WhatsApp Integration)

- Features:

o Direct communication with leads via WhatsApp chat
o Quick response system for better engagement
o Maintains conversation history

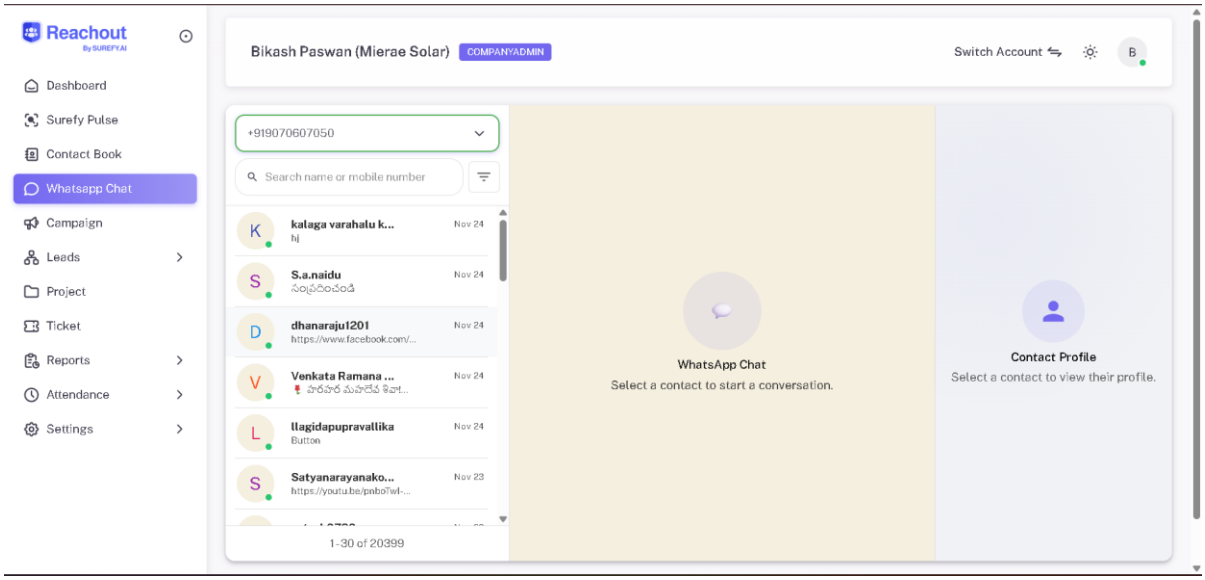


Fig . 5.1

6. Campaign Management

• Admin/User can:

o Create new campaigns using the "Add Campaign" option

o Upload and manage campaign-related files (data batches, contact lists)

o View all campaigns in a structured table with details like Created Date, Campaign Name, and Batch Count

o Search and filter campaigns using the search bar for quick access

o Edit existing campaigns using the action (edit) option

o Track number of contacts/leads associated with each campaign

6.1 Campaign -

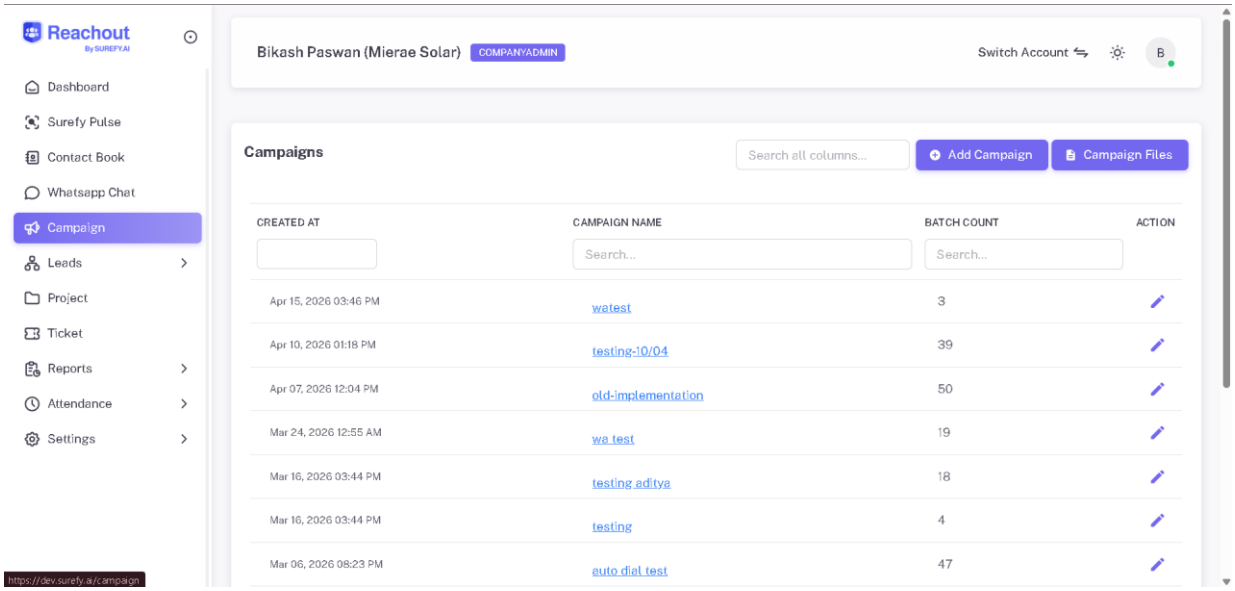


Fig . 6.1

6.2 Campaign Files -

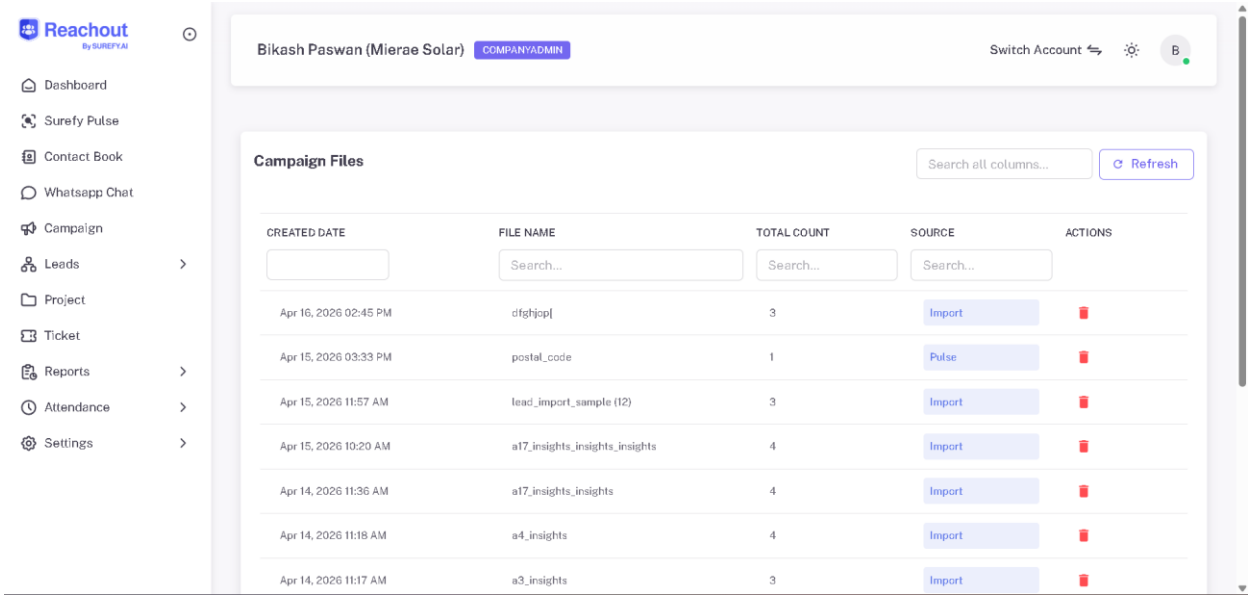


Fig . 6.2

7. Reports Managements -

• Admin/User can:

o View and manage different types of reports such as Leads, Field Staff Leads, Tasks, and Attendance

o Export data using the “New Export” option for offline use and analysis

o Refresh report data using the “Refresh” button
o Filter reports based on date range (Start Date – End Date)
o Apply status filters to view specific data
o Clear filters using the “Clear Filters” option

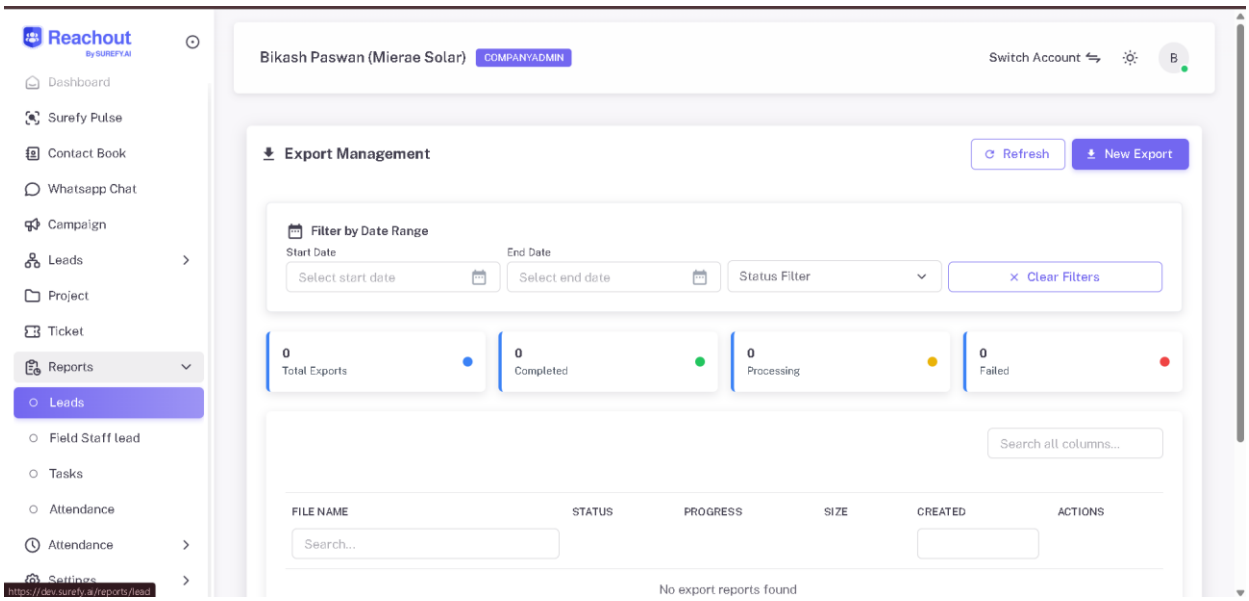


Fig . 7.1

8. Project and Tickets Managements –

• Admin/User can:
o Create and manage projects using the “Create Project” option
o Organize projects into multiple sections and subsections for structured data handling
o Add forms, fields, and file uploads within each project section
o View and manage all projects with “View All” and “Create Section” features
o Create and manage support tickets using the “Create Ticket” option
o Track ticket status such as Open, Working, On Hold, and Completed
o Assign tickets to team members and monitor their progress
o Set and manage ticket priorities (Low, Medium, High, Critical)

8.1 Projects -

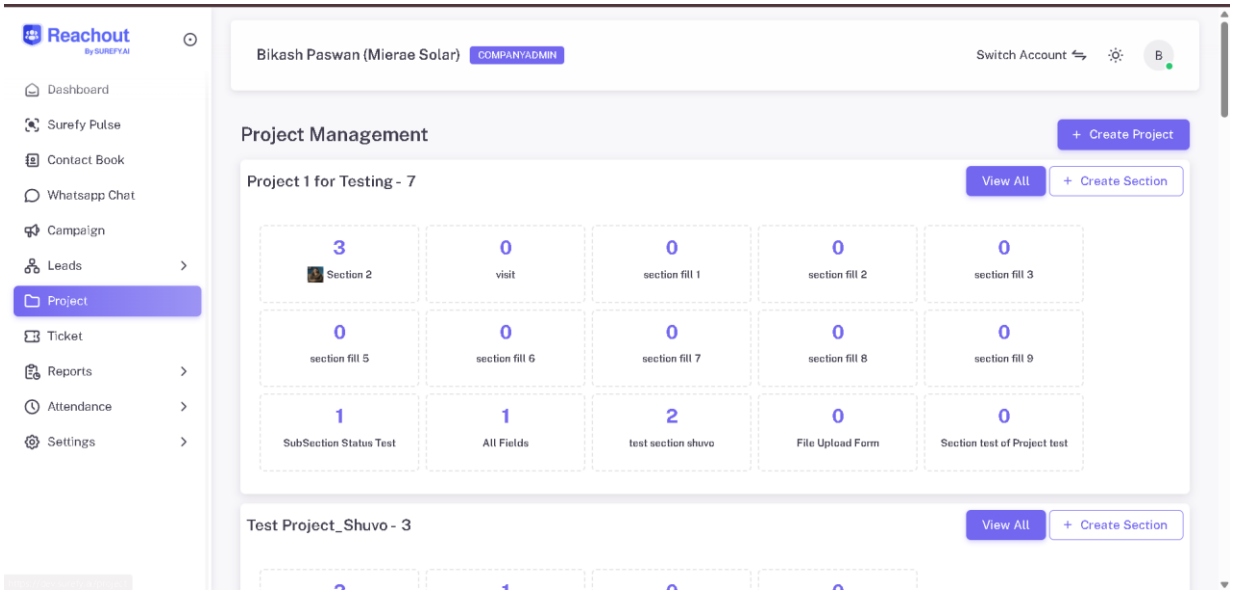


Fig . 8.1

8.2 Tickets -

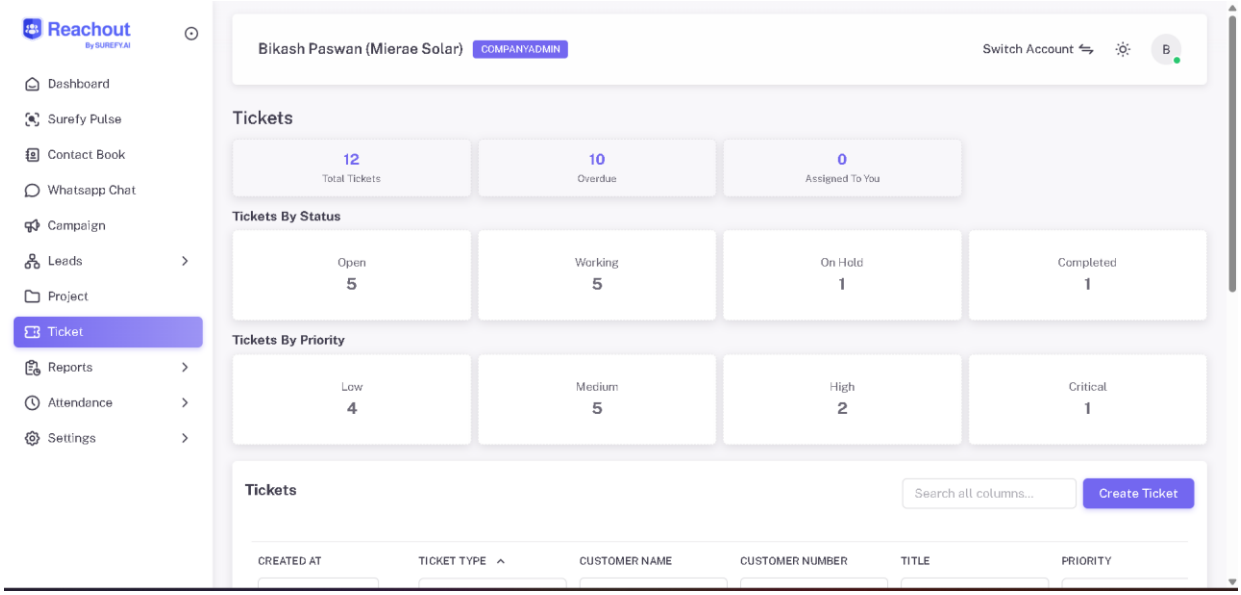


Fig . 8.2

9. Attendance Managements –

• **Admin/User can:**

o Mark attendance using **Clock In / Clock Out** functionality
o View current session details like **Clock In Time and Worked Time**
o Apply for leave using the **"Apply for Leave"** option
o Add attendance manually using **"Manual Attendance"** option
o Filter attendance records by **date range and status**
o Export attendance data for reporting purposes
o View attendance in **calendar format** for better visualization

9.1 Attendance –

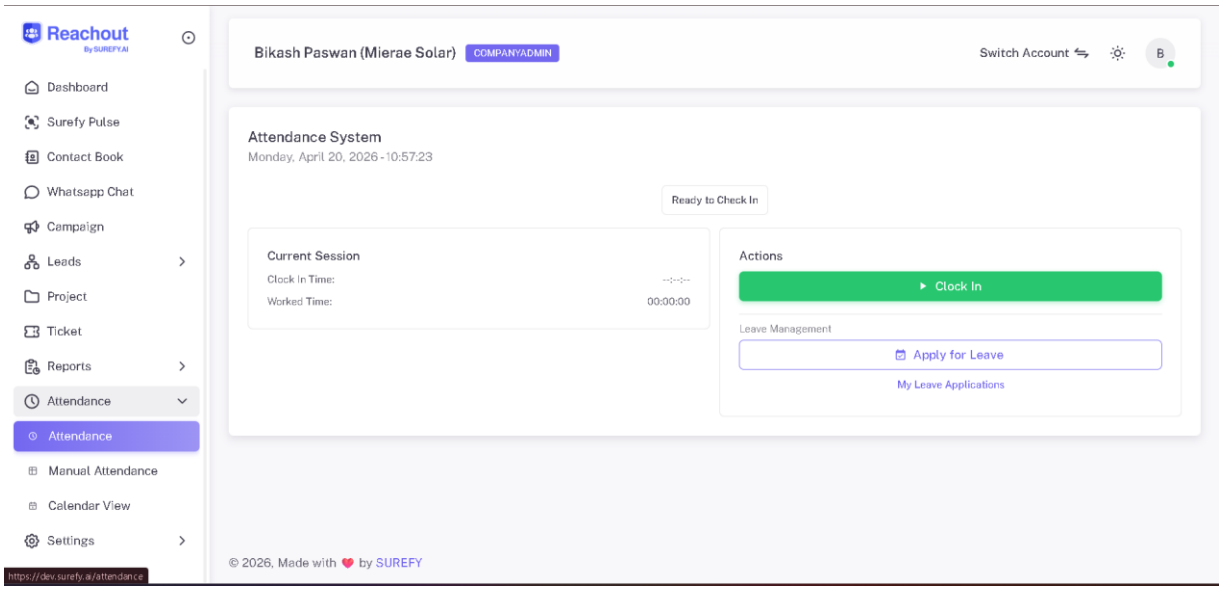


Fig . 9.1

9.2 Manual Attendance -

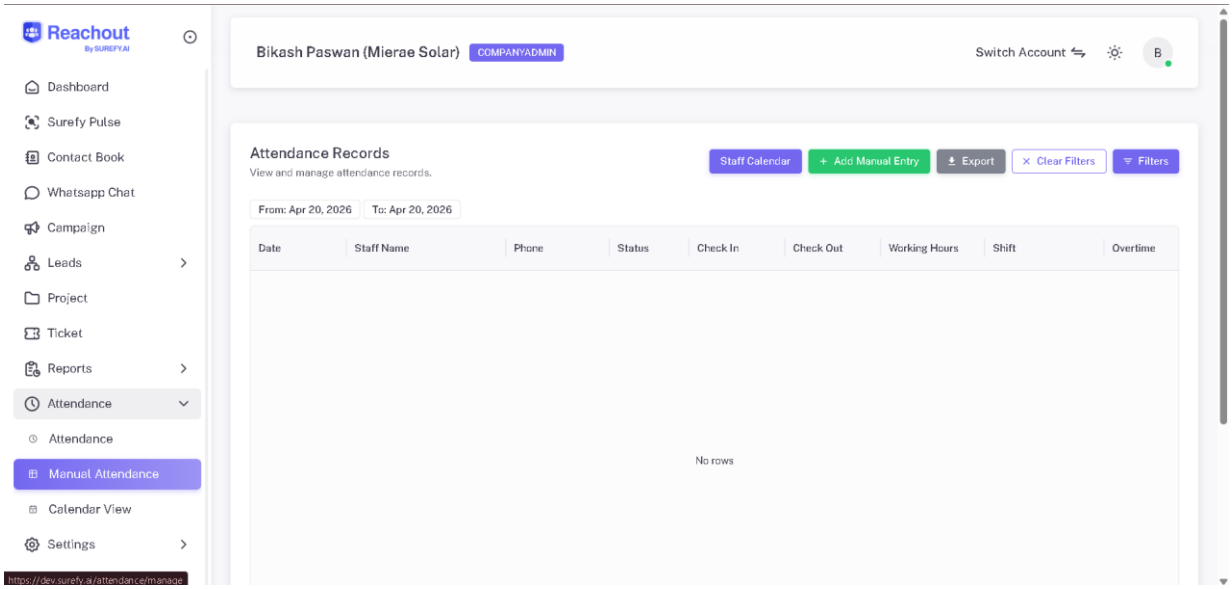


Fig . 9.2

9.3 Calendar View –

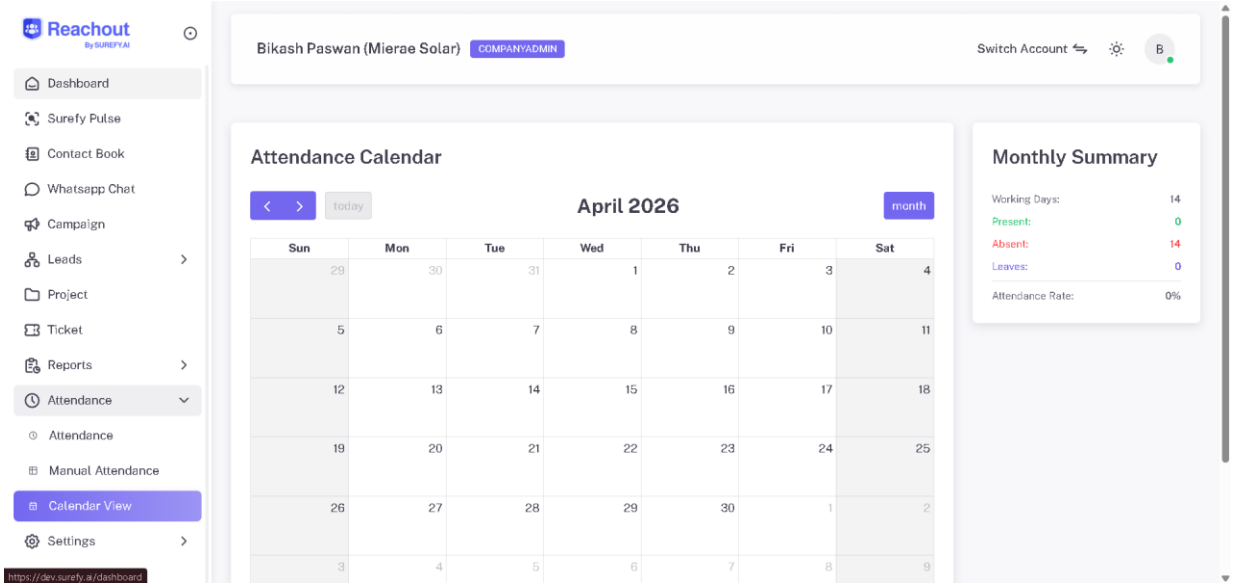


Fig . 9.3

10 Settings Management –

- o Manage system users through the Users section (add, update, assign roles)
- o Configure and manage Contact settings for better data organization
- o Customize Lead settings including lead fields and workflows
- o Manage Project configurations and related settings
- o Control Ticket system settings and workflow rules
- o Manage Product-related settings for business operations
- o Configure Communication Management (notifications, integrations, etc.)
- o Customize Column Management to control data display in tables
- o View and monitor System Logs for tracking system activities

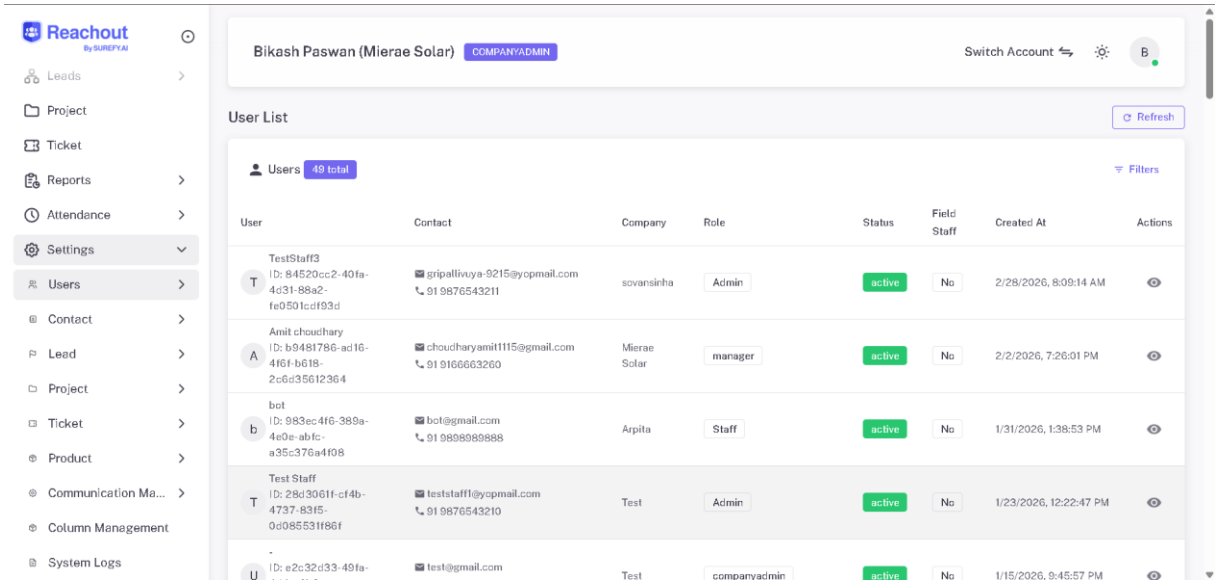


Fig . 10.1

ROLES AND RESPONSIBILITIES

During my internship as a Frontend Developer Intern, I was actively involved in developing and improving the user interface of the Surefy ReachOut platform. My role focused on building responsive components, integrating API, and ensuring smooth user experience across different modules.

1.Frontend Development

• Developed and designed UI components using Next.js (React.js)
• Worked on modules like Dashboard, Leads**,** Campaign, Reports, Attendance, and Tasks
• Ensured responsive design for different screen sizes
• Improved UI/UX for better usability and performance

2. API Integration

• Integrated backend APIs using **Axios/Fetch**
• Handled API responses and displayed dynamic data on the frontend
• Managed loading states, error handling, and data updates
• Worked closely with backend team for API structure and debugging

3. Task Management & Feature Implementation

• Completed assigned tasks related to frontend development
• Worked on **task-based system (To Do, In Progress, Done)**
• Implemented new features and improved existing functionalities
• Fixed UI bugs and optimized performance

4. State Management & Data Handling

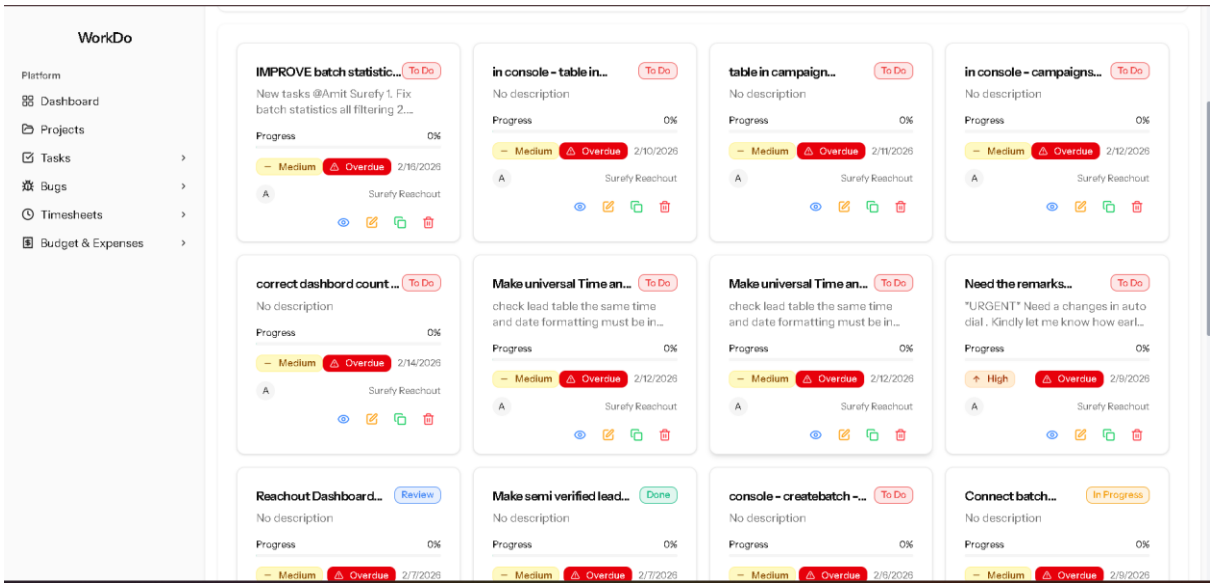
• Managed application state for smooth data flow
• Ensured real-time updates on dashboard and modules
• Handled user data, lead data, and report data efficiently

5. Collaboration & Learning

- Collaborated with team members using task management tools
- Participated in discussions and learned real-world development practices
- Improved skills in full-stack development concepts, API handling, and debugging

6. Additional Contributions

- Worked on improving **dashboard statistics and UI components**
- Assisted in testing features and identifying bugs
- Contributed to making the platform more user-friendly and efficient



OUTCOMES AND LEARINIGS

The development of **Surefy.ai ReachOut – Lead Management System** provided significant learning outcomes and practical exposure to real-world software development. This project helped in strengthening both technical and professional skills.

1. Technical Skills Gained

• Gained hands-on experience in **frontend development using Next.js (React.js)**
• Learned how to **integrate APIs** and handle real-time data
• Improved understanding of **state management and component-based architecture**
• Worked with modern tools and technologies used in industry-level applications

2. Full-Stack Development Understanding

• Understood how frontend interacts with backend systems
• Learned API structure, request-response cycle, and data flow
• Gained knowledge of **MongoDB database structure and CRUD operations**
• Experienced real-time application development and deployment concepts

3. Problem Solving and Debugging

• Improved ability to identify and fix bugs in UI and API integration
• Learned debugging techniques for handling errors and performance issues
• Developed logical thinking for optimizing features and workflows

4. Practical Industry Exposure

• Worked on a **real-world project (Surefy ReachOut platform)**
• Understood how business requirements are converted into technical solutions
• Experienced working on modules like **Dashboard, Leads, Campaigns, Reports, Attendance, and Tasks**

5. Teamwork and Collaboration

• Collaborated with team members and backend developers
• Learned task management and workflow handling in a professional environment
• Improved communication and coordination skills

6. Professional Growth

• Gained confidence in building scalable and user-friendly applications
• Improved understanding of UI/UX design principles
• Learned how to manage deadlines and complete assigned tasks efficiently

FUTURE SCOPE

The **Surefy.ai ReachOut – Lead Management System** has strong potential for further enhancement and scalability. While the current system provides efficient lead management and communication features, several improvements can be implemented to make it more powerful and intelligent.

1. AI-Based Lead Analysis

• Integration of AI/ML models to analyze lead behavior
• Predict lead conversion probability (Hot/Warm/Cold automatically)
• Provide smart suggestions for follow-ups and communication

2. Advanced Automation

• Automated follow-up messages via WhatsApp, Email, and SMS
• Workflow automation for lead assignment and status updates
• Scheduled campaigns and reminders

3. Mobile Application Development

• Development of Android/iOS mobile apps for better accessibility
• Real-time notifications for leads, tasks, and updates
• Improved usability for field staff and sales teams

4. Enhanced Reporting & Analytics

• Advanced dashboards with charts and visual insights
• Custom report generation based on business needs
• Performance tracking using KPIs and trends

5. Third-Party Integrations

• Integration with CRM tools, payment gateways, and marketing platforms
• Email marketing tools (like Mailchimp) integration
• Calendar integration for scheduling meetings and follow-ups

6. Improved Security Features

• Multi-factor authentication (MFA)
• Data encryption for sensitive information
• Advanced role-based access control and audit logs

7. Scalability and Cloud Optimization

• Deployment on scalable cloud infrastructure (AWS, Azure, etc.)
• Microservices architecture for handling large-scale users
• Performance optimization for high traffic

CONCLUSION

The development of **Surefy.ai ReachOut – Lead Management System** has been a highly valuable and insightful experience. It provided a strong understanding of how to design, develop, and implement a real-world web application focused on business automation and lead management. From building responsive user interfaces to integrating APIs and managing dynamic data, this project offered hands-on exposure to modern frontend and full-stack development practices.

Surefy ReachOut effectively addresses the challenges faced in traditional lead management systems, such as manual tracking, scattered communication, and inefficient follow-ups. By providing features like centralized dashboard, lead tracking, campaign management, communication tools, reporting, and role-based access, the platform simplifies business workflows and enhances productivity and decision-making.

Throughout the development process, I strengthened my technical skills in **Next.js (React), API integration, and frontend development**, while also improving my problem-solving, debugging, and system understanding abilities. Working on real-time features and handling live data helped me gain practical experience in building scalable and user-friendly applications.

In conclusion, Surefy ReachOut is not just a project but a practical solution for modern businesses to manage leads and improve customer engagement efficiently. This experience has enhanced my confidence in developing industry-level applications and prepared me for future challenges in the field of software development.